Question 1: To be answered in (10) Minutes [ ] / 2 Marks

Choose one correct answer for each statement and fill the following table:

Question	1	2	3	4
Answer				

1. What is the closest time complexity of the function $f(n) = n \log n + n$
   a. $O(n)$
   b. $O(\log(n!))$
   c. $O(n^2)$
   d. $O(\log n)$

2. If $f(n) = n^3 \log n + n$, then:
   a. $f(n) = \Omega(n!)$
   b. $f(n) = \Omega(\log n)$
   c. $f(n) = \Omega(n^4)$
   d. *All the above*

3. If $f(n) = 2^{n+1}$ and $g(n) = 2^n$ then $f(n)$ is:
   a. $O(g(n))$
   b. $\Omega(g(n))$
   c. $\Theta(g(n))$
   d. *All the above*

4. If $f(n) = n^2$, then:
   a. $f(n) = o(n^3)$
   b. $f(n) = o(n \log n)$
   c. $f(n) = o(n^2)$
   d. $f(n) = o(n \log(n^2))$

Student Name (in English): _____ Student ID: _____

Question 2: To be answered in (20) Minutes [ ] / 3 Ma

a) What is the time complexity of the following algorithm in the most precise Big-O? (1 point)

```
Sort(List) {
i = 0
  While (i< List.Length) {
    SmallestElement = List[i]
        j= I +1
    While (j < List.Length) {
      if(SmallestElement > List[j]) {
        SmallestElement = List[j]
      }
        j = j+1
    }
    Swap(List[i], SmallestElement)
        i = i+1
  }
}
```

b) Prove that running time $T(n) = n^3 + 20n + 1$ is not $O(n^2)$. (2 points)